Topic 3 — Energy, Heat and Thermal Science

Current Affairs (2024–2026 Focus)

- **70th BPSC (Pre 2024) Heat Transfer Without Medium Fact**:
- **Thermal Radiation** is the only mode of heat transfer that does **NOT require any material medium** (propagates via electromagnetic waves across vacuum, e.g. solar energy reaching Earth). Conduction and convection strictly require material media.
- **70th BPSC (Pre Re-Exam 2024) Steam Burn Severity Fact**:
- Burns caused by steam at 100°C are far more severe and dangerous than burns from boiling water or air at 100°C due to the **high latent heat of vaporization ($540\ \text{cal/g} = 22.6 \times 10^5\ \text{J/kg}$)** released during condensation on the skin.
- **MP PCS (Pre 2023) Ecosystem Energy Degradation Fact**:
- The progressive decline of energy as it passes from lower to higher trophic levels (only $\approx 10\%$ passed; Lindeman's Rule) is fundamentally governed by the **Second Law of Thermodynamics** (spontaneous processes increase entropy, accompanied by dissipation of metabolic heat).
- **69th BPSC (Pre 2023) Official Heat Wave Temperature Criteria**:
- IMD declares a Heat Wave when the maximum temperature reaches at least 40°C **or more for Plains** and at least 30°C **or more for Hilly Regions**.
- **67th BPSC (Pre 2022) Triple Point of Water Fact**:
- The unique thermodynamic state where solid, liquid, and gaseous water coexist in simultaneous equilibrium is the **Triple Point of Water**: $273.16\ \text{K} = 0.01^\circ\text{C} = 32.018^\circ\text{F}$ at a partial vapor pressure of $611.657\ \text{Pa} = 6.11657\ \text{mbar} \approx 0.0060366\ \text{atm}$.
- **UPPCS (Pre 2025) Sweating & Humidity Fact**:
- Evaporation of sweat from human skin is an **endothermic physical process** that absorbs latent heat from subcutaneous blood, dissipating body heat. High ambient relative humidity decreases the skin-air vapor pressure gradient, suppressing evaporation and causing acute discomfort.
- **UPPCS (Pre 2024) Humidity & Temperature Interdependence**:
- **Relative humidity decreases** as air temperature rises (warm air has greater moisture-holding capacity). **Absolute humidity increases** with increasing surface evaporation.
- **National Green Hydrogen Mission:** India targets **5 MMT (Million Metric Tonnes)** per annum green hydrogen production capacity by **2030**, supported by **125 GW** of associated renewable energy capacity. Reduces ₹1 lakh crore worth of fossil fuel imports. Hydrogen has the **highest calorific value** among all known chemical fuels (~$150\ \text{kJ/g}$ or $150\ \text{MJ/kg}$), producing only water vapor on combustion.
- **PM Surya Ghar: Muft Bijli Yojana (February 2024):** Aims to install rooftop solar power systems in **1 crore (10 million) households**, providing up to **300 units of free electricity** per month with a total financial outlay of ₹75,021 crore.
- **COP28 UAE Consensus & India's Target:** Tripling global renewable energy capacity by 2030. India's updated NDC (*Panchamrit* targets announced at Glasgow COP26) commits to achieving **500 GW of non-fossil energy capacity by 2030** and reducing emissions intensity of GDP by **45%** by 2030 relative to 2005 levels.
- **India's First 100% Solar-Powered Airport: Cochin International Airport (CIAL)**, Kerala became the world's first fully solar-powered airport in August 2015, winning the UN Champions of the Earth Award.
- **Ultra-Supercritical Thermal Power Units:** NTPC's deployment of ultra-supercritical coal units operating at pressures above $22.1\ \text{MPa}$ and temperatures exceeding 600°C, elevating Carnot thermal efficiency to $> 42\%$, thereby reducing specific coal consumption and CO_2 emissions.
- **ITER & Tokamak Fusion Reactor Progress:** International Thermonuclear Experimental Reactor (Cadarache, Southern France). Magnetic confinement fusion utilizing deuterium-tritium plasma heated to $> 150\ \text{million}\ ^\circ\text{C}$ (ten times hotter than the Sun's core). India is one of the seven key founding members providing the cryostat (built by L&T).
- **Cryogenic Upper Stage (C32 / CE-20) in Gaganyaan & LVM3:** Liquid hydrogen fuel stored at -253°C ($20\ \text{K}$) and liquid oxygen oxidizer stored at -183°C ($90\ \text{K}$), delivering a very high specific impulse ($I_{sp} \approx 443\ \text{s}$).
- **Energy Conservation Building Code (ECBC) & Cool Roofs:** Implementing high Solar Reflectance Index (SRI) paints and phase-change building materials to block solar thermal infrared radiation, curbing indoor temperatures by $2\text{–}5^\circ\text{C}$.

3.1 Work, Energy and Power

Work & Work-Energy Theorem

- **Mathematical Definition:** Work done by a constant force is the scalar product (dot product) of force and displacement vectors:

$$W = \vec{F} \cdot \vec{d} = Fd\cos\theta$$

- **Positive Work ($0^\circ \le \theta < 90^\circ$):** Force has a component in the direction of displacement (e.g., gravity doing work on a falling stone).
- **Negative Work ($90^\circ < \theta \le 180^\circ$):** Force opposes motion (e.g., kinetic friction acting on a sliding vehicle, work done by brakes).
- **Zero Work ($\theta = 90^\circ$ or $d = 0$):**
  - A coolie carrying luggage on his head and walking on a horizontal railway platform does **zero work against gravity** because displacement is horizontal and gravitational force is vertically downward ($\theta = 90^\circ$).
  - Centripetal force acting on the Moon orbiting the Earth does **zero work** because centripetal force is always directed radially toward the center, perpendicular to orbital velocity.
- **Work-Energy Theorem:** The net work done by all forces acting on a body equals the change in its kinetic energy:

$$W_{\text{net}} = \Delta KE = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

Forms of Energy & Momentum Link

- **Kinetic Energy (KE):** Energy possessed by a body by virtue of its motion: $KE = \frac{1}{2}mv^2$.
- Expressed in terms of linear momentum ($p = mv$):

$$KE = \frac{p^2}{2m} \implies p = \sqrt{2m \cdot KE}$$

- **Must-Score Calculation Facts:**
  - If momentum is doubled ($p' = 2p$), kinetic energy becomes **4 times** ($KE' = 4KE$, a 300% increase).
  - If momentum increases by 50%, KE increases by $(1.5)^2 - 1 = 2.25 - 1 = 125\%$.
  - If kinetic energy is increased by 300%, momentum increases by $\sqrt{4} - 1 = 100\%$.
- **Potential Energy (PE):** Energy stored by virtue of position or configuration:
- **Gravitational Potential Energy:** $PE = mgh$ (for heights $h \ll R_{\text{earth}}$).
- **Elastic Potential Energy (Spring):** $PE = \frac{1}{2}kx^2$, where k is the spring constant and x is displacement.
- **Law of Conservation of Mechanical Energy:** In an isolated conservative system, total mechanical energy ($E = KE + PE$) remains strictly constant:

$$\frac{1}{2}mv^2 + mgh = \text{constant}$$

Power & Units

- **Power (P):** Rate of doing work: $P = \frac{dW}{dt} = \vec{F} \cdot \vec{v}$.
- **SI Unit:** Watt ($1\ \text{W} = 1\ \text{J/s} = 1\ \text{kg} \cdot \text{m}^2/\text{s}^3$).
- **Horsepower (HP):** Imperial engineering unit: $1\ \text{HP} = 746\ \text{Watts}$.
- **Kilowatt-hour (kWh):** Commercial unit of energy:

$$1\ \text{kWh} = 1000\ \text{W} \times 3600\ \text{s} = 3.6 \times 10^6\ \text{J} = 3.6\ \text{MJ}$$

3.2 Energy Resources: Conventional, Non-Conventional and Renewable Transition

Classification Matrix

1. **Conventional Sources:** Sources in commercial use for generations: - Fossil fuels (Coal, Petroleum, Natural Gas). - Large Hydroelectric Power (thermal/mechanical conversion).
2. **Non-Conventional Sources:** Sources harnessed commercially only in modern times: - Solar energy, Wind energy, Tidal energy, Geothermal energy, OTEC (Ocean Thermal Energy Conversion), Biomass, Nuclear energy.
3. **Renewable Sources:** Naturally replenished at a rate equal to or faster than consumption: - Solar, Wind, Hydro, Biomass, Geothermal, Tidal, Wave.
4. **Non-Renewable Sources:** Finite geological reserves exhausted with use: - Coal, Crude Oil, Natural Gas, **Nuclear fuels (Uranium, Thorium)**.

Master Comparison of Fuel Calorific Values

- The calorific value of a fuel is the quantity of heat produced by complete combustion of unit mass ($1\ \text{kg}$ or $1\ \text{g}$) of that fuel.

$$\text{Hydrogen } (150\ \text{kJ/g}) > \text{LPG } (50\ \text{kJ/g}) \approx \text{CNG/Methane } (50\ \text{kJ/g}) > \text{Petrol/Diesel } (45\ \text{kJ/g}) > \text{Kerosene } (43\ \text{kJ/g}) > \text{Anthracite Coal } (33\ \text{kJ/g}) > \text{W}$$

Renewable Energy Technologies

- **Solar Photovoltaic (PV):** Direct conversion of sunlight into direct current (DC) electricity via the **photovoltaic effect** in semiconductor silicon cells (bandgap $\sim 1.1\ \text{eV}$).
- **Solar Thermal:** Parabolic mirrors or central receiver towers concentrate solar flux to vaporize fluids (molten salts/steam) to drive conventional turbines.
- **Wind Power:** Kinetic energy of wind drives aerodynamic blades: $P_{\text{wind}} = \frac{1}{2}\rho A v^3$ (Power is proportional to the **cube of wind speed** v^3). Betz limit dictates theoretical maximum efficiency is 59.3%.
- **Ocean Thermal Energy Conversion (OTEC):** Exploits the temperature difference between warm surface tropical ocean waters ($25°\text{C}$) and cold deep ocean waters ($5°\text{C}$). Requires a minimum temperature gradient of $20°\text{C}$ to operate low-boiling working fluids like ammonia.
- **Geothermal Energy:** Taps underground steam and hot water reservoirs heated by magma. In India, prominent geothermal fields include **Puga Valley (Ladakh)**, Manikaran (Himachal Pradesh), and Tatapani (Chhattisgarh).

3.3 Heat, Internal Energy and Thermometry

Heat vs Internal Energy

- **Internal Energy (U):** The total microscopic kinetic energy (translational, rotational, vibrational) plus potential energy (intermolecular intermolecular forces) of all molecules in a thermodynamic system.
- **Heat (Q):** Thermal energy transferred across a system boundary solely due to a **temperature difference**. Heat is **energy in transit**; a body *possesses* internal energy, but *transfers* heat.
- **Mechanical Equivalent of Heat (J):** Discovered by Joule: $W = JQ$, where $J = 4.184\ \text{J/cal}$.

Thermometric Scales & Interconversions

- Temperature scales are defined using two standard fixed points: the lower fixed point (ice melting point) and the upper fixed point (steam boiling point).

$$\frac{C-0}{100}=\frac{F-32}{180}=\frac{K-273.15}{100}=\frac{R-0}{80}$$

Simplifying:

$$\frac{C}{5}=\frac{F-32}{9}=\frac{K-273.15}{5}$$

- **Critical Must-Score Checkpoints:**
- -40° **Point:** Celsius and Fahrenheit scales read the exact same numerical value:

$$\frac{C}{5}=\frac{C-32}{9}\implies 9C=5C-160\implies 4C=-160\implies C=-40^\circ$$

  Hence, $-40^\circ\text{C}=-40^\circ\text{F}$.
- 574.25° **Point:** Fahrenheit and Kelvin scales read the same value: $574.25^\circ\text{F}=574.25\ \text{K}$.
- **Normal Human Body Temperature:**

$$37.0^\circ\text{C}=98.6^\circ\text{F}=310.15\ \text{K}$$

- **Degree Size:** $1\ \text{K}=1^\circ\text{C}$. A temperature rise of 10°C is equivalent to a temperature rise of $10\ \text{K}$, but corresponds to a rise of 18°F ($10\times\frac{9}{5}$).

3.4 Specific Heat Capacity, Calorimetry and Latent Heat

Specific Heat Capacity (c)

- The amount of heat required to raise the temperature of unit mass ($1\ \text{kg}$ or $1\ \text{g}$) of a substance by 1°C ($1\ \text{K}$):

$$Q=mc\Delta T\implies c=\frac{Q}{m\Delta T}$$

- **Molar Heat Capacity (C_m):** Heat required per mole: $C_m=\frac{Q}{n\Delta T}$.
- For gases: $C_p-C_v=R$ (**Mayer's Relation**).
- Specific heat ratio: $\gamma=C_p/C_v$ (Monatomic $\gamma=1.67$, Diatomic $\gamma=1.40$, Triatomic $\gamma=1.33$).
- **Water's Thermal Anomaly:** Liquid water has the highest specific heat capacity among all common solids and liquids:

$$c_{\text{water}}=1.0\ \text{cal/g}\cdot^\circ\text{C}=4184\ \text{J/kg}\cdot\text{K}\approx 4200\ \text{J/kg}\cdot\text{K}$$

- **Engine Cooling:** Water circulates around cylinders absorbing massive engine combustion heat without boiling.
- **Hot Water Bottles:** Releases sustained heat slowly over hours due to high heat storage capacity.
- **Climatic Equalizer:** Oceans buffer maritime land from extreme seasonal swings.

Calorimetry & Principle of Mixtures

- In an insulated container (calorimeter), neglecting heat loss to the environment:

$$\text{Heat Lost by Hotter Body}=\text{Heat Gained by Colder Body}$$

$$m_1c_1(T_1-T_{\text{final}})=m_2c_2(T_{\text{final}}-T_2)$$

Latent Heat (L)

- Heat absorbed or released during a phase change occurring at **constant temperature and constant pressure**:

$$Q=mL\implies L=\frac{Q}{m}$$

- **Latent Heat of Fusion (L_f):** Solid $\leftrightarrow$ Liquid transition. For ice:

$$L_f=80\ \text{cal/g}=3.34\times10^5\ \text{J/kg}=334\ \text{kJ/kg}$$

- **Latent Heat of Vaporization (L_v):** Liquid $\leftrightarrow$ Gas transition. For water:

$$L_v=540\ \text{cal/g}=2.26\times10^6\ \text{J/kg}=2260\ \text{kJ/kg}$$

- **Physical Reason $L_v\gg L_f$:** In melting, crystal lattice bonds are merely loosened while molecules remain close. In boiling, intermolecular hydrogen bonds are completely severed and molecules are pulled far apart against atmospheric pressure, requiring massive energy ($540\ \text{cal/g}$ vs $80\ \text{cal/g}$).

3.5 Thermal Expansion and Anomalous Expansion of Water

Coefficients of Thermal Expansion

1. **Linear Expansion (α):**

$$\Delta L = L_0 \alpha \Delta T \implies \alpha = \frac{\Delta L}{L_0 \Delta T} \quad (\text{Unit: } \mathrm{K^{-1}} \text{ or } \mathrm{^\circ C^{-1}})$$

2. **Superficial / Areal Expansion (β):**

$$\Delta A = A_0 \beta \Delta T \quad (\beta \approx 2\alpha)$$

3. **Volumetric / Cubical Expansion (γ):**

$$\Delta V = V_0 \gamma \Delta T \quad (\gamma \approx 3\alpha)$$

- **Master Ratio for Isotropic Solids:**

$$\alpha : \beta : \gamma = 1 : 2 : 3$$

Practical Manifestations & Must-Score Engineering Facts

- **Railway Track Expansion Gaps:** Rails expand in summer. Without deliberate gaps, compressive thermal stress ($\sigma = Y\alpha\Delta T$) buckles the tracks.
- **Bimetallic Strips:** Made of brass ($\alpha = 19 \times 10^{-6}\ \mathrm{K^{-1}}$) and iron/invar ($\alpha = 12 \times 10^{-6}\ \mathrm{K^{-1}}$). Heating causes the strip to curve with brass on the outside, breaking electric circuits in iron thermostats.
- **Fitting Iron Rims on Wooden Cart Wheels:** The iron rim is made slightly smaller in diameter than the wooden wheel. It is heated until it expands, slipped onto the wheel, and rapidly quenched with cold water. On contracting, it grips the wheel firmly.
- **Pendulum Clock Errors:**

$$T = 2\pi\sqrt{\frac{L}{g}} \implies \frac{\Delta T}{T} = \frac{1}{2}\frac{\Delta L}{L} = \frac{1}{2}\alpha\Delta\theta$$

- Summer (temperature increases): L increases $\implies$ T increases $\implies$ clock loses time (runs slow).
- Winter (temperature decreases): L decreases $\implies$ T decreases $\implies$ clock gains time (runs fast).
- Modern fix: Using **Invar** (alloy of 36% Nickel and 64% Steel) with near-zero thermal expansion ($\alpha \approx 1.2 \times 10^{-6}\ \mathrm{K^{-1}}$).

Anomalous Expansion of Water

- Most liquids expand uniformly upon heating. Water behaves abnormally between $0^\circ\mathrm{C}$ **and** $4^\circ\mathrm{C}$:
- When water at $0^\circ\mathrm{C}$ is heated, its volume **decreases** (contracts) until it reaches $4^\circ\mathrm{C}$ **(**$3.98^\circ\mathrm{C}$**)**.
- Above $4^\circ\mathrm{C}$, it expands normally.
- Consequently, water has **minimum volume** and **maximum density (**$1.000\ \mathrm{g/cm^3} = 1000\ \mathrm{kg/m^3}$**)** at $4^\circ\mathrm{C}$.
- **Molecular Origin:** In ice, hydrogen bonds create an open hexagonal cage-like crystalline lattice with large empty interstitial voids (making ice less dense than water, $\rho_{\text{ice}} \approx 0.917\ \mathrm{g/cm^3}$). Upon melting at $0^\circ\mathrm{C}$, this open cage structure collapses, packing water molecules closer together up to $4^\circ\mathrm{C}$. Above $4^\circ\mathrm{C}$, standard thermal agitation dominates and expands the fluid.
- **Survival of Aquatic Organisms in Frozen Lakes:**
- As winter temperatures drop, surface water cools toward $4^\circ\mathrm{C}$, becomes denser, and sinks to the bottom.
- Once the entire water body reaches $4^\circ\mathrm{C}$, further cooling of surface water below $4^\circ\mathrm{C}$ makes it *less dense*, keeping it at the surface.
- Surface water reaches $0^\circ\mathrm{C}$ and freezes into an ice crust.
- Ice is a poor conductor of heat and floats at the top, shielding the water below.
- The water at the lake bottom remains liquid at $+4^\circ\mathrm{C}$, preserving aquatic animal life throughout the harshest winter.

3.6 Modes of Heat Transfer: Conduction, Convection and Radiation

1. Conduction

- **Mechanism:** Transfer of heat through matter without any macroscopic translation of the material itself. Vibrating lattice atoms pass kinetic energy to neighbors; in metals, free conduction electrons accelerate energy transfer.
- **Fourier's Law of Heat Conduction:**

$$\frac{dQ}{dt} = -kA\frac{dT}{dx}$$

where k is the **thermal conductivity**, A is cross-sectional area, and $\frac{dT}{dx}$ is the temperature gradient.

- **Thermal Conductivities Hierarchy:**

$$\text{Silver } (429) > \text{Copper } (401) > \text{Gold } (317) > \text{Aluminum } (237) > \text{Iron } (80) > \text{Glass } (0.8) > \text{Water } (0.6) > \text{Wood } (0.13) > \text{Air } (0.026)$$

- **Everyday Conduction Phenomena:**
- **Metal vs Wooden Chair:** In cold winter, sitting on a metal chair feels much colder than sitting on a wooden chair at the same temperature because metal has high thermal conductivity and rapidly draws heat away from your skin.
- **Cooking Utensil Bottoms:** Cooking pans are often made of stainless steel with a copper-clad base because copper conducts heat five times faster than steel, preventing hot-spot burning.
- **Ice Preservation in Sawdust:** Sawdust and gunny bags contain millions of trapped air pockets. Because air is a poor conductor, heat cannot penetrate to melt the ice.

2. Convection

- **Mechanism:** Heat transfer through fluids (liquids and gases) via the **actual bulk motion of fluid particles** driven by density variations under gravity.
- **Buoyancy & Convection Currents:** Fluid near the heat source absorbs heat, expands, becomes less dense, and ascends. Cooler, denser fluid descends under gravity to replace it, establishing a closed circulation loop.
- **Absolute Requirement of Gravity:** Convection is entirely impossible in a weightless environment (orbiting spacecraft). Without buoyant force ($F_b = V\rho g$), warm fluid cannot rise. In space stations, air must be mechanically circulated by electric fans to prevent astronauts from suffocating in their own exhaled CO_2.
- **Geophysical Applications:**
- **Land and Sea Breezes:** Driven by diurnal differential heating rates of land and sea.
- **Atmospheric Trade Winds:** Solar heating at the equator causes warm air to ascend and flow toward the poles, while polar air moves toward the equator, deflected by Coriolis force.
- **Mantle Convection:** Radioactive heat within the Earth's mantle drives slow convection currents in semi-molten rock, driving plate tectonics and continental drift.

3. Radiation

- **Mechanism:** Heat transfer via electromagnetic waves (chiefly infrared photons). Requires **no material medium** and propagates across interstellar vacuum at the speed of light ($c = 3 \times 10^8\ \text{m/s}$).
- **Absorptive and Reflective Properties:**
- Highly polished, white, or silvered surfaces reflect radiant heat and absorb very little.
- Rough, dark, or matte black surfaces absorb almost all incident radiant energy.
- White clothes are worn in summer because they reflect solar radiation, keeping the body cool; dark clothes are preferred in winter to absorb warmth.
- **The Thermos (Dewar) Flask Design:**
- Invented by Sir James Dewar (1892) to store cryogenic liquefied gases.
- **Double-walled glass vessel:** High structural strength and poor conduction.
- **Vacuum between walls:** Completely eliminates heat transfer by **conduction** and **convection**.
- **Silvered internal glass surfaces:** Silver mirror coating reflects thermal radiation back, stopping heat loss or gain by **radiation**.
- **Insulating stopper & support pad:** Cork/plastic lid blocks air convection currents at the opening.

3.7 Laws of Thermal Radiation and Atmospheric Meteorology

Fundamental Radiation Laws

1. **Prevost's Theory of Heat Exchange:** All bodies at any temperature above absolute zero ($T > 0\ \text{K}$) continuously radiate thermal energy to their surroundings and simultaneously absorb radiation from their surroundings. Thermal equilibrium simply means rate of emission equals rate of absorption.
2. **Kirchhoff's Law of Radiation:**

$$\frac{e_\lambda}{a_\lambda} = E_{\text{black}}$$

   At any temperature, the ratio of emissive power to absorptive power for a given wavelength is constant for all bodies and equal to the emissive power of an ideal black body. - **Must-Score Corollary: Good absorbers are good emitters.** A black body absorbs all incident radiation without reflection or transmission; when heated, it radiates maximum energy. - **Solar Fraunhofer Lines:** Dark absorption lines in the solar spectrum arise because cooler gases in the solar atmosphere absorb specific wavelengths that they emit when heated.
3. **Stefan-Boltzmann Law:**

$$E = \sigma T^4$$

   The total radiant energy emitted per second per unit area of a black body is directly proportional to the fourth power of its absolute thermodynamic temperature (T). - Constant $\sigma = 5.670 \times 10^{-8}\ \text{W/m}^2 \cdot \text{K}^4$. - For a non-black body with emissivity e ($0 < e < 1$): $E = e\sigma T^4$. - Net radiant loss to surroundings at temperature T_0: $E_{\text{net}} = e\sigma(T^4 - T_0^4)$. - *Must-Score Problem:* If the temperature of a hot filament increases from $200\ \text{K}$ to $400\ \text{K}$ (doubles), its rate of energy emission multiplies by $2^4 = 16\ \text{times}$.
4. **Wien's Displacement Law:**

$$\lambda_{\text{max}} \cdot T = b$$

   where $b = 2.898 \times 10^{-3}\ \text{m} \cdot \text{K}$ (Wien's constant). - As a body gets hotter, the peak of its radiation spectrum shifts toward **shorter wavelengths** (higher frequencies). - A heating iron poker first glows dull red, then bright orange, yellow, and finally brilliant white-blue as temperature climbs.
5. **Newton's Law of Cooling:**

$$\frac{dT}{dt} = -k(T - T_0)$$

   For small temperature differences between a body (T) and its ambient surroundings (T_0), the rate of heat loss by cooling is directly proportional to the temperature excess $(T - T_0)$.

Atmospheric Heat Trapping & Meteorology

- **Greenhouse Effect:** Solar short-wave radiation (visible and near-UV) penetrates through the atmosphere and heats the Earth's surface. The warm Earth re-radiates energy as **long-wave infrared radiation**. Atmospheric greenhouse gases (CO_2, H_2O vapor, CH_4, N_2O, O_3, CFCs) absorb this outgoing

terrestrial infrared radiation and re-emit it in all directions, warming the lower troposphere by $\approx 33^\circ\text{C}$ (natural average temperature $+15^\circ\text{C}$ instead of -18°C).

- **Why Cloudy Nights are Warmer:** Dense low cloud cover acts as a physical infrared blanket. Outgoing terrestrial heat radiated by the ground is absorbed and back-radiated by water droplets in the clouds, preventing the ground from cooling down. On clear, starry nights, heat escapes directly into outer space, producing sharp nighttime temperature drops.
- **Dew & Frost Formation Conditions:**
- **Dew:** Liquid condensation formed on grass and leaves when surface temperature falls below the **Dew Point** (the temperature at which relative humidity of air reaches 100% and saturation occurs). Favorable conditions: **clear sky, calm air (no wind), high relative humidity, long winter night**.
- **Frost:** Formed when the dew point lies **below 0°C (below freezing point)**; water vapor undergoes direct deposition into ice crystals without forming liquid droplets.
- **Fog vs Smog:**
- **Fog:** Cloud at ground level composed of suspended microscopic water droplets, reducing visibility below $1\ \text{km}$.
- **Smog:** Smoke + Fog. Industrial/vehicular air pollutants (SO_2, NO_x, unburnt hydrocarbons) act as condensation nuclei, forming toxic persistent haze. **Photochemical smog** is oxidizing in nature and rich in ozone (O_3) and PAN (peroxyacetyl nitrate).

3.8 Thermodynamics, Heat Engines, Refrigeration and Phase Transitions

Thermodynamic Systems and Processes

- **Zeroth Law:** Introduces **Temperature** as the property that governs thermal equilibrium.
- **First Law ($\Delta Q = \Delta U + W$):** Energy conservation. Internal energy (U) is a state function depending only on initial and final states, whereas heat (Q) and work (W) are path functions.
- **Four Classical Thermodynamic Processes:** 1. **Isothermal ($\Delta T = 0$):** Temperature remains constant. Internal energy $\Delta U = 0 \implies Q = W$. Governed by Boyle's law ($PV = \text{const}$). Work done: $W = nRT\ln(V_2/V_1)$. Must occur **infinitely slowly in containers with perfectly conducting walls**. 2. **Adiabatic ($Q = 0$):** Zero heat exchange with surroundings. Work is done entirely at the expense of internal energy: $W = -\Delta U$. Governed by $PV^\gamma = \text{constant}$ (where $\gamma = C_p/C_v$). Occurs **very rapidly in thermally insulated containers**.
  - *Practical Examples:* Sudden bursting of an automobile tire leads to rapid expansion against atmospheric pressure, causing noticeable chilling/cooling; sudden compression of air in a bicycle pump heats the barrel. 3. **Isobaric ($\Delta P = 0$):** Pressure remains constant. Work done $W = P(V_2 - V_1) = nR\Delta T$. Heat added: $Q = nC_p\Delta T$. 4. **Isochoric / Isometric ($\Delta V = 0$):** Volume remains constant. Work done $W = 0$. Heat added goes solely into internal energy: $Q = \Delta U = nC_v\Delta T$.

Second Law of Thermodynamics & Heat Engines

- **Clausius Statement:** It is impossible to construct a cyclic machine whose sole effect is the transfer of heat from a cooler body to a hotter body without external work.
- **Kelvin-Planck Statement:** It is impossible to construct a cyclic engine that absorbs heat from a single thermal reservoir and converts it completely into mechanical work without rejecting waste heat.
- **Carnot Engine:** Theoretical ideal reversible heat engine operating between a hot source (T_1) and cold sink (T_2).
- Efficiency of Carnot Engine:

$$\eta = \frac{W}{Q_1} = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1} = \frac{T_1 - T_2}{T_1}$$

- η depends **only on the temperatures of the source and sink**, completely independent of the working substance.
- Efficiency is always strictly $< 100\%$ (1.0) because sink temperature T_2 cannot be absolute zero ($0\ \text{K}$).

Refrigerators, Air Conditioners & The Open Door Trap

- **Working Principle:** A refrigerator is a reverse heat engine. The working fluid (refrigerant) absorbs heat Q_2 from the internal cold food compartment at temperature T_2, electric motor does work W on the compressor, and heat $Q_1 = Q_2 + W$ is rejected to the room via external condenser coils at temperature T_1.
- **Coefficient of Performance (COP):**

$$\beta = \frac{Q_2}{W} = \frac{T_2}{T_1 - T_2}$$

- **The Classic Open Door Trap:**
- If a domestic refrigerator is left running with its door wide open in a completely closed, sealed room, **the temperature of the room will steadily increase (warm up)**.
- *Reason:* The heat discharged into the room by the condenser coils on the rear is equal to the heat absorbed from the interior plus the electrical energy consumed by the compressor motor ($Q_1 = Q_2 + W$). Because $Q_1 > Q_2$, net positive heat is pumped into the room air.

Bilingual Terminology -- Energy, Heat and Thermal Physics

English Term	Hindi Term	Key Anchor
Energy (ऊर्जा)	ऊर्जा	Capacity to do work; SI unit = Joule (J)

English Term	Hindi Term	Key Anchor
Kinetic Energy (गतिज ऊर्जा)	गतिज ऊर्जा	Energy of motion; KE = 1/2 mv2
Potential Energy (स्थितिज ऊर्जा)	स्थितिज ऊर्जा	Stored energy due to position; PE = mgh (gravitational)
Conservation of Energy (ऊर्जा संरक्षण)	ऊर्जा संरक्षण का नियम	Energy cannot be created or destroyed; only transforms
Heat (ऊष्मा)	ऊष्मा	Form of energy; transferred from hot to cold body
Temperature (ताप)	ताप	Measure of hotness/coldness; determines direction of heat flow
Thermodynamics (ऊष्मागतिकी)	ऊष्मागतिकी	Science of heat-work relationships
Specific Heat Capacity (विशिष्ट ऊष्मा धारिता)	विशिष्ट ऊष्माधारिता	Heat needed to raise 1 kg substance by 1 K; water = 4200 J/kg/K
Latent Heat (गुप्त ऊष्मा)	गुप्त ऊष्मा	Heat absorbed/released during phase change without temperature change
Conduction (चालन)	चालन	Heat transfer through matter by vibration; metals = good conductors
Convection (संवहन)	संवहन	Heat transfer by movement of fluid/gas; sea breeze, boiling water
Radiation (विकिरण)	विकिरण	Heat transfer through electromagnetic waves; no medium needed; e.g., sunlight
Greenhouse Effect (हरित गृह प्रभाव)	हरित गृह प्रभाव	Trapping of Earth's infrared radiation by greenhouse gases
Absolute Zero (परम शून्य)	परम शून्य	0 Kelvin = -273.15 degC; minimum possible temperature
Entropy (एन्ट्रॉपी)	एन्ट्रॉपी	Measure of disorder; increases in all natural processes (2nd Law of Thermodynamics)
Thermostat (ताप नियंत्रक)	ताप-स्थायी	Device that regulates temperature automatically

One-Liner Revision -- Energy, Heat and Thermal Physics

#	Fast Fact
1	1 Joule = 1 Newton x 1 metre (work done)
2	1 Calorie = **4.186 Joules** (heat to raise 1 g water by 1 degC)
3	1 kWh = 3.6 x 10^6 J = 3.6 MJ (unit on electricity bill)
4	KE = 1/2 mv2; PE = mgh; Total ME = KE + PE = constant (no friction)
5	Power = work done / time = Energy / time; SI unit = Watt (W)
6	1 Watt = 1 Joule/second; 1 kW = 1000 W; 1 MW = 10^6 W
7	1 Horsepower = **746 Watts**; British HP used in motors/vehicles
8	Water has the HIGHEST specific heat (4200 J/kg/K); best coolant and heat reservoir
9	Specific heat of water = 4200 J/kg/K; iron = 450 J/kg/K; lead = 128 J/kg/K
10	Latent heat of vaporisation of water = 2260 kJ/kg (much higher than fusion = 336 kJ/kg)
11	0 degC = 273.15 K; 100 degC = 373.15 K; -273.15 degC = 0 K (absolute zero)
12	Celsius to Fahrenheit: F = (9/5)C + 32; at -40 degC = -40 degF
13	Heat transfer: Conduction (solids) > Convection (fluids) > Radiation (vacuum/space)
14	Metals = good conductors; Wood, glass, air = insulators; vacuum = no conduction/convection
15	Thermos flask = prevents heat loss by all 3 methods (vacuum, silvered walls, stopper)
16	Black surfaces = absorb AND emit heat best (best radiator and absorber)
17	Stefan-Boltzmann Law: E = sigma x T^4 (radiation proportional to 4th power of temp)
18	Newton's Law of Cooling: rate of cooling proportional to (T_object - T_surrounding)
19	Greenhouses: glass transmits visible light but blocks infrared (re-radiated heat)
20	Greenhouse gases: CO2, CH4, N2O, H2O vapour, O3, CFCs
21	Most potent GHG (per molecule) = **SF6** (sulfur hexafluoride); most volume-significant = CO2
22	Global warming potential (GWP): CH4 = 25x CO2; N2O = 298x CO2 (100-yr)
23	1st Law of Thermodynamics = Conservation of Energy (dU = Q - W)
24	2nd Law of Thermodynamics = entropy of universe always increases; heat flows hot to cold
25	3rd Law of Thermodynamics = entropy of perfect crystal = 0 at absolute zero
26	Zeroth Law = if A and B are in thermal equilibrium with C, then A and B are in equilibrium

#	Fast Fact
27	Thermal expansion: gases > liquids > solids (rail gaps, bimetallic strips)
28	Bimetallic strip = two metals with different expansion rates; used in thermostats
29	Regelation = ice melts under pressure and refreezes when pressure released (ice skating)
30	Clinical thermometer range = 35 degC to 42 degC; mercury (now digital)